Table of Contents

Main Notebook

Season Goals

Members Present	EDP Steps	Problems or Goals	Scheduling
All Members	Team Management Time Management Identify the Problem	Determine goals for the season to set priorities.	Completed Date: Apr 25, 2026 - Apr 25, 2026

Team Goals

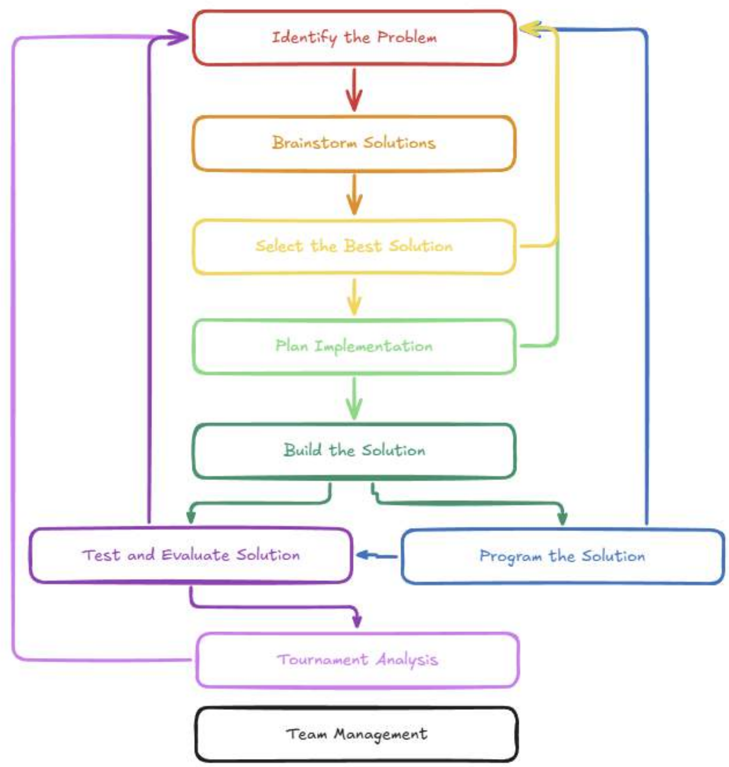
To help us determine our priorities for design and overall decisions, we've compiled a list of season-wide goals to aim for. We will address whether and when we achieve a goal, and set specific per-tournament goals as well.

- **Win a tournament** - Most members of our team have never won a tournament, so we want to achieve that this season.
- **Win three excellence awards** - Our team has won an excellence award in the past, and if we can improve our documentation even more, we think we can win even more this season.
- **Keep consistent documentation.** In the past, we have not been the best at keeping up with documentation consistently. We want this notebook to be written with the brainstorming and construction of this robot.
- **Make SF at a Signature Event** - Signature Events are near the highest level of competition, so we want to try to make it far at one of them.
- **Have a top 10 skills score in Indiana** - One thing that we have learned is that skills are the only thing we can really control. You could get unlucky qualification matches, alliances could get scorched, and judges could interpret notebooks differently. Skills scores are your team's score, regardless of what happens.
- **Have the highest programming score in Indiana**. It is our programmers' last season, and we want to prioritize skills, so this is a good goal that brings them together. We can achieve this by dedicating time to programming and designing the robot to be efficient at skills and tournament matches.
- **Maintain a top 150 skills score in the World throughout the season:** As seen in previous seasons, the roll-down on the top 10 skills unqualified in the world to receive qualifications normally goes slightly further than #150.
- **Qualify for the state** - This is our ERC, and we need to win an Excellence Award or a Tournament Champion Award (at local competitions) to qualify for it. It has many opportunities to qualify for the worlds tournament through placement and awards.
- **Win a judged award at state (Amaze, Think, Innovate, Design, or Excellence)** - This would qualify us for the world championship. We have won a judges' award at the state level in the past, and these awards are all a tier higher than the judges' award.
- **Qualify for the worlds** - This is the highest level tournament of the season, and is the final tournament of the season.

Our Design Process

Members Present	EDP Steps	Problems or Goals	Scheduling
All Members	Team Management Iterating	Explain our team's design process and how we interpret each step.	Completed Date: Apr 25, 2026 - Apr 25, 2026

EDP Flowchart



A flowchart of our design process. Made by Liam 7701S, 4-25-26.

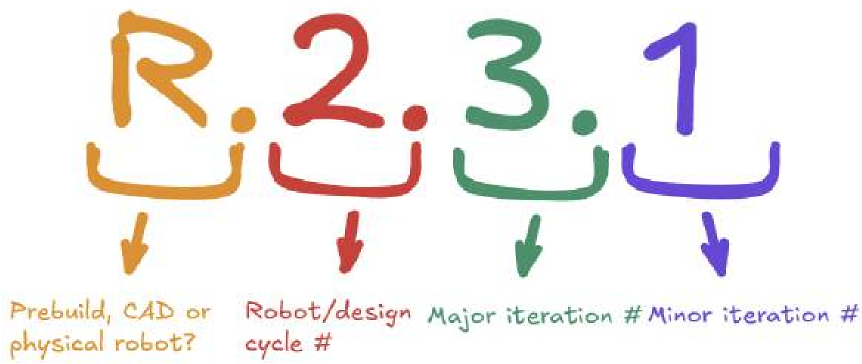
- We start with identifying the problem, which could include game/tournament analysis, trying to solve an issue with the robot/program, etc.
- We then brainstorm possible solutions. Some could be our own, or some could be based on outside research/testing. We want to try to understand each idea fully, which could mean running tests or considering pros and cons.
- Afterwards, we select the best solution. This should usually be quantitative, like the result of a test or by using a decision matrix. We will always try to explain our decision fully.
- Plan implementation is when we get ready to build. This could be done by explaining the CAD process or constructing prototypes to test how things function.
- Building the solution is when we actually get to building the robot. It will be a step-by-step explanation of how we reached our final product.
- After building, we either go to testing to see how well things work or programming if they seem to work well. Many of the steps have arrows leading back to older steps. This is because the design process is iterative, and we need to go back and solve future problems.
- Finally, after testing and evaluation, we go to tournament analysis. This will be after each tournament, where we will explain what went well, what did not go to plan, and what problems we need to solve. There is also a step for team management, but it is floating in space because it is not dependent on the design process, but rather on what is happening with our team/club.

Color Code

Each step has a color and a place in the header under "EDP Steps" (Identify the Problem). This should help us understand what steps we are going through during each entry. There are also additional steps (ex, Budgeting) which don't fit directly into the design process but are still worth recognizing.

Numbering System - Ex: R.2.3.1

Members Present	EDP Steps	Problems or Goals	Scheduling
Liam Shipman	Team Management	Explain our system for numbering our robot iterations.	Completed Date: Apr 25, 2026 - Apr 25, 2026



An example of numbering. A real one can be found in the title (see L.#.#.# above). By 7701S, 4/25/26.

Explanation

Letter: Is the entry or picture referring to the prebuild (P) (planning, brainstorming before building the robot/mechanism), the CAD (C), or the physical robot/build (R)?

- In this example, we are referring to the physical build.

First number: What full design cycle is this referring to (complete rebuild)?

- In this example, this is the second robot of the season.

Second number: What major design iteration is this? (ex, completely rebuilding a subsystem, adding/removing a mechanism)

- In this example, there have been 3 major iterations.

Third Number: What minor design iteration is this? (ex, modifying a mechanism, adding something small)

- In this example, there has been one minor iteration.

When we increase a number by one, succeeding numbers reset to zero (ex, P.2.1.4-> P.3.0.0, R.1.3.4->R.1.4.0).

Why?

This numbering system will allow us to keep track of our current and past design iterations, which will help with referencing photos, previous designs, and help judges understand what design we are talking about. We have used it before, and we have found it to be very useful.

Credit

This format was created and used by our team in the Push Back Season. The image of the page from our notebook can be found in Appendix C, and the link to our old notebook can be found in Appendix A.

Pre-Season Budget and Event Plans

Members Present	EDP Steps	Problems or Goals	Scheduling
Liam Shipman Nolan Wolff Sam Wilson	Team Mana... Budgeting	Explain the budget makeup for our team over the season, discuss the plan for events, and find what time each member will be free.	In Progress Date: Apr 26, 2026 - Apr 26, 2026

Budgeting

Income

Our team has two main sources of money to spend: club funds and personal funds.

- Club funds are what our school provides us with, which (based on the 2025-26 season, as we do not yet know the true budget for the next season) is $500. These funds can be spent on parts that our team specifically needs. The club also pays for standard events, but does not come out of our team's funds.
- Personal funds are the money our team members/parents spend. This can be used for parts, but it can also be used for custom apparel, stickers, etc. We also must spend personal funds for the Signature Event and Worlds registration fees. Personal funds are not a set amount, and we will most likely spend as we need.

Inventory

We also have separate inventories of parts. The parts that our club owns are free for members of any team to use, but each team is guaranteed an electronics kit (brain, batteries, etc.) and two pneumatic kits. We also have a large inventory of personal parts, which our team members have personally purchased and accumulated over time.

Inventory of Sam's parts

Event Plans

As of right now, no events are published on RobotEvents yet, so we don't know when or where each event is. Once they get released, we will look through and plan all our events. Also, we would like to compete at the Mall of America signature event, so we need to have a robot ready by the start of August.

Game Analysis

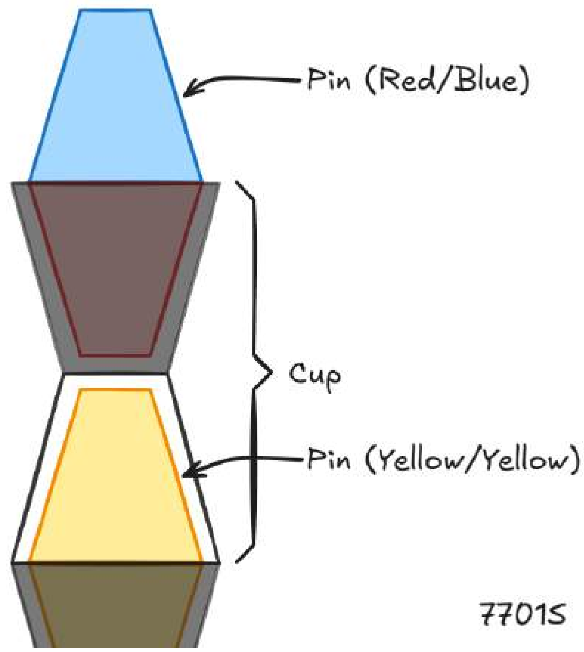
Members Present	EDP Steps	Problems or Goals	Scheduling
All Members	Identify the Pr... Game Analysis	Discuss & analyze the game to find what problems a robot must solve.	In Progress - N/A Date: Apr 27, 2026 - May 1, 2026

General Overview

This year's game, Override, is a challenge where robots need to create stacks of cups and pins. Each pin has two sides, with different colors corresponding to different alliances. There are four combinations: red/blue, red/yellow, blue/yellow, and yellow/yellow. You score pins by putting them in Goals or nesting them between cups. Cups are hourglass-shaped and can fit half of a pin on each side. Also, one side is transparent, and one is opaque. Whichever color of pin is visible is counted for points.

There are three types of goals: alliance, neutral, and center-neutral. They all have different height dimensions.

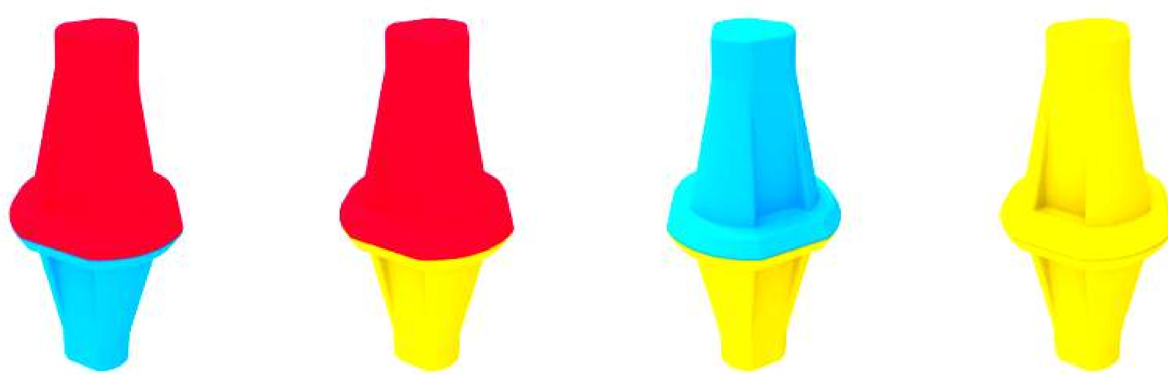
In this sketch, the blue and yellow are scored because they are visible.



A sketch of pins and cups, 7701S, 4/27/26.

Pins

- Each pin has two halves of different colors (combinations are red/blue, red/yellow, blue/yellow, and yellow/yellow).



Showing the different pin combinations from the Override Game Manual

- 63 total, 4 red/blue, 20 red/yellow, 20 blue/yellow, 19 yellow/yellow. Each alliance gets ten preloads of their red or blue/yellow pins and one yellow/yellow pin.
- Each robot gets a pin as a preload <SG5>.
- 6.5” tall and about 3” wide.

- A pin must be fully nested in a goal or a cup that is part of a stack to be considered scored <SC2>.
- A pin must be visible for that color to be scored (not covered by the cup's opaque side) <SC3>.
- A yellow pin can be owned by a team using a toggle <SC3>, <SC4>, <SC5>.
- Start in many different locations and may be matchloaded as well <SG11>.
- An alliance-colored pin that is scored (not covered by an opaque cup) is worth 5 points to that alliance.
- A yellow pin is worth 0 points by default, but is worth 10 points if owned by an alliance using a toggle <SC3>, <SC4>, <SC5>.

Cups

- Used to score pins, it has an opaque and a transparent side.
- 6.5" tall and about 3" wide.
- Hourglass shape has two open sides that pins can fit into.
- Start in multiple locations, including as matchloads <SG11>.
- Can be matchloaded with or without a pin <SG11>.
- All cups are identical.
- 63 total, 10 match loads per team, 24 on the field with the gray side up, and 12 with the gray side down.

A picture of a cup, from the Override Game Manual

Design Criteria (Pins and Cups)

- Our robot needs to pick up pins from different orientations (horizontal/verical on the field, in a cup, in the matchloader).
- Our robot needs to be able to rotate pins so you can control which side of the pins is scored.
- Our robot needs to be able to pick up cups from multiple orientations.
- Our robot needs to be able to rotate cups to control which side of a pin is scored.
- Our robot needs to be able to collect a pin/cup stack.
- Our robot needs to be able to lift pins to put them into a cup/goal.

Goals

- 3 different sizes: alliance, neutral, and the central goals.
- Each goal has a hole to fit a pin inside and an identifiable April Tag.
- There are four alliance goals, two for each alliance. They are 3.25" tall.
- There are four neutral goals. They are 5.77" tall.
- There is an additional goal in the center of the field (in what is known as the Midfield) which is 8.77" tall.
- You can stack as many cups and pins as you want.

A picture of the different goals, from the Override Game Manual

Design Criteria (Goals)

- Our robot needs to be able to identify each goal with the AprilTags.
- Our robot needs to be able to lift pins and cups into each goal.
- Our robot needs to be able to lift pins and cups to make large stacks at variable heights.

Toggles

- Four around the field, one in each quadrant.
- Triangular prism, each side has a different color.
- Starting at yellow, when rotated to be red or blue, the red/blue alliance takes ownership of each yellow pin in the quadrant <SC4>, <SC5>.
- Ownership makes each yellow pin in a quadrant worth 10 points.
- On a linear slide, you can move them up/down.
- Similar to Spin Up rollers.

A picture of a toggle, from the Override Game Manual

Design Criteria (Toggles)

- Our robot needs to change the orientation of the toggle.
- Our robot should be able to do that quickly, as it could result in large point swings.
- Our robot should be able to move quickly between quadrants because of the high-point swings with toggle control.
- Our robot should be able to detect the color of a toggle.
- Toggles are located on the field perimeter, so your toggle mechanism must reach that height.

Loaders

- Similar to Push Back Loaders, four of them, two per alliance on each Alliance Station perimeter wall.
- You can load a pin, a cup, or a pin/cup stack <SG11>.
- Loaders may also be lifted by a drive team member to load a pin nested inside a cup.
- 3.25" opening at the bottom when not lifted, 11.85" opening when lifted.
- Each alliance gets 11 pin matchloads and 10 cup matchloads.

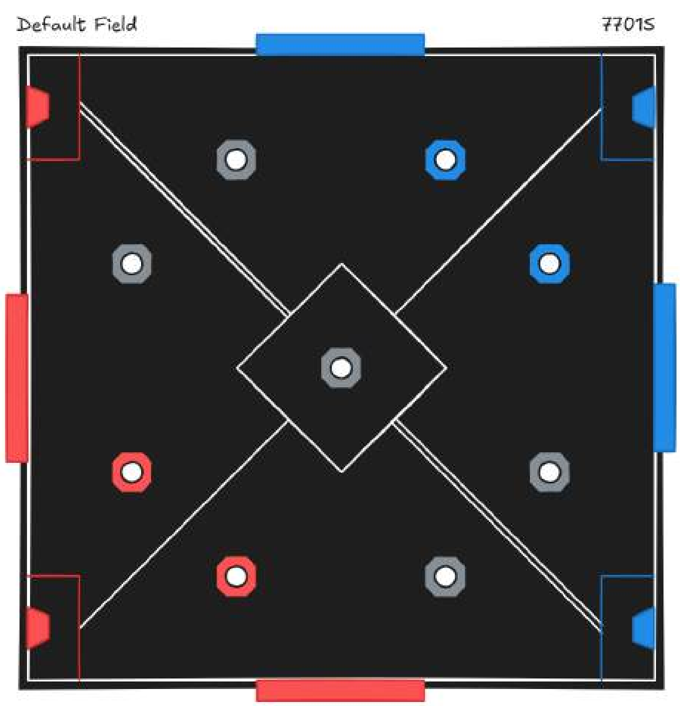
A picture of a loader, from the Override Game Manual

Design Criteria (Loaders)

- Our robot must be able to remove any combination of pins and cups from the loader.
- Our robot must be able to remove pins from the loader when open or closed.
- Due to the high amount of match loads, our robot should be able to do this very quickly.

Gameplay Rules

- Scoring is evaluated after the match <SC1>.
- You cannot descore any pin/cup that is scored on a goal (except on alliance goals) <SG10>.
- You can only score on a neutral goal or your alliance goal <SG9>.
- Possession is limited to only one pin and one cup at a time <SG6>.
- Don't remove scoring objects from the field <SG4>.
- At the end of the match, each robot in the midfield gets 8 points, and the alliance with the majority of robots in the midfield gains ownership of yellow pins in the midfield <SC5b>, <SC6>.
- To get an Autonomous Win Point, our alliance must score at least 2 pins on three different goals and must have 7 pins owned for your alliance (alliance color or use toggles to control yellow ones).



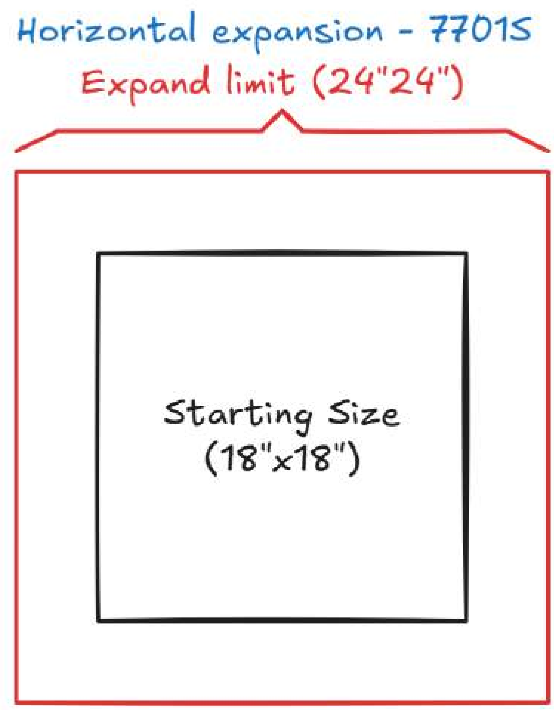
A sketch showing the field without pins or cups, made by 7701S 4/28/26.

Design Criteria (Gameplay)

- Our robot needs to be designed to build stacks, not take down the opponent's stacks. If you violate this twice in a match or three times in different matches, it is a major violation. Also, if it is match-affecting (within 15 points), it is a major violation as well.
- Our robot needs to effectively stack one at a time and won't knock down a tower when scoring.
- Our robot needs to be fast so we can quickly move between quadrants and get to the midfield at the end of the match.
- Our robot needs to be able to push other robots out of the midfield to stop them from getting bonus points.
- Our robot needs to be able to quickly switch between scoring cups and pins to adjust depending on the strategy and what is already scored.

Significant Robot Rules/Constraints

- Our robot must start the match within an 18” cube <R3>.
- Our robot may expand horizontally to be 24”x24” <SG2>.
- Our robot may expand vertically to 50” <SG3>.
- If you are in the midfield during the endgame (last 20 seconds), your robot can not exceed 18” vertically <SG12>.
- We may only use parts on the official legal parts list or commercially available screws, nuts, standoffs, etc that fit into the legal 8-32 screws <R16>, <R22>.
- Only one robot is allowed <R1>.
- Our subsystem 1 (“Mobile robotic base including wheels, tracks, legs, or any other mechanism that allows the Robot to navigate the majority of the flat playing Field surface”) may not exceed a motor power of 55 watts (made of 11-watt and 5.5-watt motors) <R11>.



- Our robot's motor power cannot exceed 88 watts (made of 11-watt and 5.5-watt motors) <R10>.
- Our robot cannot have parts or mechanisms that could damage the field, robots, or cause entanglement, and may not have 3D-printed parts <R18>.
- We may moderately use lubricant, hot glue cable connections, use commercial cable-management wrappings, use commercial zip ties or rubber bands within specific sizes, cool motors with aerosol-based freeze-spray for cooling motors, and use any commercial string ¼” or less <R19>.
- We cannot change the color of any part via dyeing, painting, or anodizing <R23a>.
- We can have up to 12 pieces of custom plastic, each piece not being bigger than 4”x8”. “Legal plastic types are polycarbonate (Lexan), acetal monopolymer (Delrin), acetal

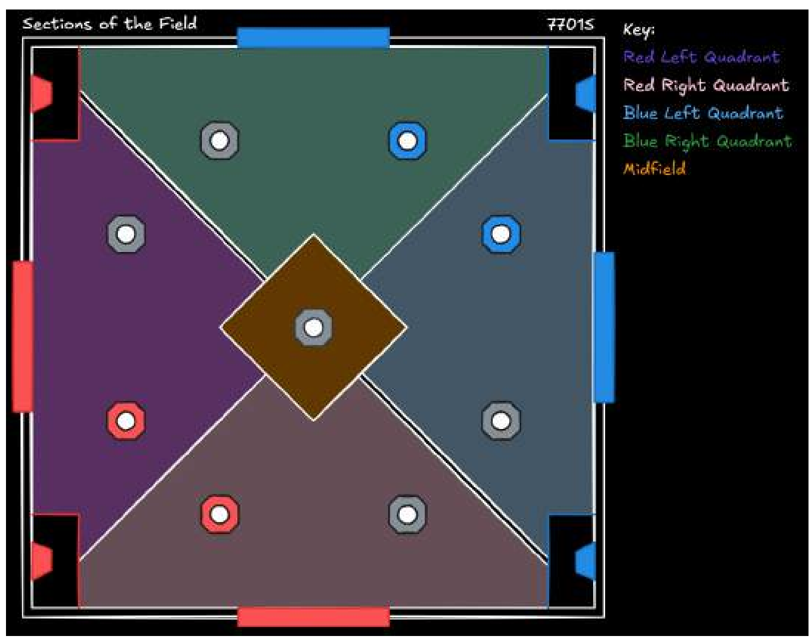
copolymer (Acetron GP), POM (acetal), ABS, PEEK, PET, HDPE, LDPE, nylon (all grades), polypropylene, PTFE, and FEP” <R24>.
- We may only use two pneumatic air tanks (no limit to the amount of pistons/solenoids) and must use an air gauge <R25>, <R26>.
- We may physically modify non-electronic plastic or metal parts <R27>.

Potential Challenges

- The main challenge presented with this is the size limitations, as we cannot exceed 24”x24” and must start 18”x18”. This means we must design our robot to be very compact.
- Another challenge is that we can only use a limited amount of custom plastic pieces, so we need to use them only when crucial and very sparingly.
- We need to find an effective drivetrain rpm because of the new 55-watt limit. It should be able to move quickly and precisely while also supporting the weight of a large lifting mechanism.
- Another robot challenge is with weight, as having more makes your drivetrain less efficient. To mitigate weight, we can buy lighter legal parts (as compared to the official VEX parts) like aluminium nylocks as permitted in <R22>.

Field Observations

- The field is rotationally symmetrical (the right side for the blue alliance is the same as the right for the red alliance) along a diagonal autonomous line.
- Each alliance's portion of the field has four goals, two alliances, and two neutrals.
- The neutral goals already start with a yellow pin.
- There are many cups pre-placed with yellow/yellow or red/blue pins across the field.
- The horizontal pins seem to be much more complicated to pick up and index into your robot.
- Each quadrant has access to 5 cups with pins, two goals, 6 empty cups, and 4 pins lying horizontally.



A sketch showing the different zones of the field, made by 7701S 2/28/26.

- We must start touching the perimeter of the field in our alliance's zone, not contacting any scoring or field elements, limiting possible locations <SG1>.

Autonomous Period

- 15-second period at the start of a match.
- No driver input allowed.
- Work with your team to score more than the opponent to receive the autonomous bonus of 12 points.

- We can also work with our alliance to gain an Autonomous Win Point (score at least 2 pins on three different goals and must have 7 pins owned for your alliance).
- AWP is important because it is the first category when determining qualification rankings (you get two for winning a match, one for tying, and one for completing the AWP tasks).
- Our robot needs to be fast and precise to do as much as possible during the period consistently.
- We also need to have multiple routes planned and coded to fit our alliance's needs.

Initial Autonomous Idea

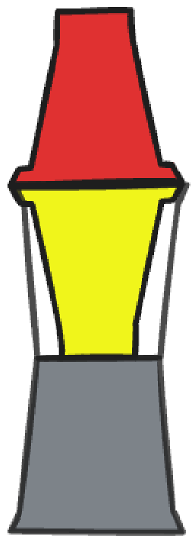
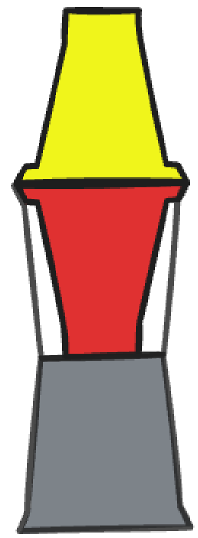
Start by taking ownership of a quadrant with a toggle. Then, place your preload on your adjacent alliance goal. Pick up and place a cup with a yellow/yellow on top of the alliance goal. Then, score another cup/pin on the neutral goal (that already has a yellow/yellow pin on it). This could be effective as it only requires 2.5 cycles (you start with your preload) to score 4 pins. It also does not require you to pick up horizontal pins, and you only need to be able to collect a cup/pin stack. This would score 40 points if done correctly and would be very close to meeting the AWP criteria.

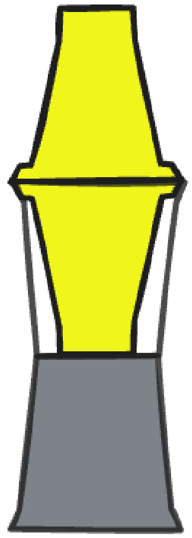
Scoring Strategy Comparison

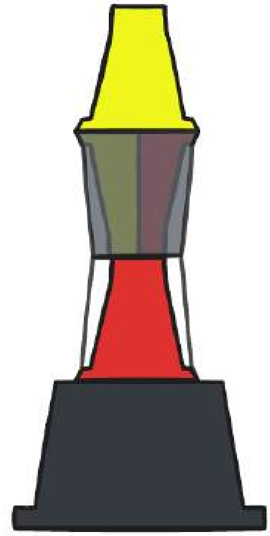
In past games, we have done an overall point breakdown to see where all the points would come from in an ideal scenario. However, due to the high number of scoring objects and the near-infinite field combinations, this would be extremely difficult.

Instead, we will directly compare different stack combinations to help understand potential strategies. We will compare different Red/Yellow stacks (because those are the match loads) to see if scoring taller or faster would be better. We will also think through the implementation of that scoring strategy, such as how hard the stack is to score, how long it takes to assemble, and how difficult and time-consuming it would be to get the scoring elements necessary to make it.

	RYO 	YRO 
Description	One red scored	One yellow scored
Max Points	**5 points**	**10 points**
Potential	**0 points**	**10 points**, **0 points**

Outcomes		
Pros	Fast, can be gotten from match load.	Fast, can be gotten from match load.
Cons	No guaranteed points, not many points.	No guaranteed points, relies on a toggle for points.
Other Notes	Protects what it is scored on.	Protects what it is scored on.
	RYC 	YRC 
Description	One yellow and red scored	One red and yellow scored
Max Points	**15 points**	**15 points**
Potential Outcomes	**5 points** & **10 points**, **5 points** & **0 points**	**5 points** & **10 points**, **5 points** & **0 points**
Pros	Fast, can be gotten from match load.	Fast, can be gotten from match load, guaranteed points, top pin getting covered doesn’t lose us points.
Cons	Not many points, many ways to take away our points	Not many points without red toggle, blue can score more than us with a toggle.
Othen Notes	Descores the pin it is scored on.	Protects the pin it’s scored on
	YYC	YYO

		
Description	Two yellows scored	One yellow scored
Max Points	**20 points**	**10 points**
Potential Outcomes	**20 points**, **0 points**	**10 points**, **0 points**
Pros	Faster, 4 of this stack can be found along a diagonal of the field, high potential points	Faster, 8 of these can be found around the perimeter of the field.
Cons	No guaranteed points, all points rely on toggles	No guaranteed points, all points rely on toggles
	RYOYG 	RYORG 
Description	A red and a yellow scored	Two reds scored
Max Points	**15 points**	**10 points**
Potential Outcomes	**10 points**, **10 points** & **5 points**, **0 points**	**5 points**
Pros	Potentially higher scoring	Guaranteed points

Cons	Slightly riskier with the yellow pin, top pin could be covered	Not as high scoring, top pin could be covered
	YYOYG / YROYG 	YYORG / YRORG 
Description	Two yellows scored	A yellow and a red scored
Max Points	**20 points**	**15 points**
Potential Outcomes	**20 points**, **10 points**, **10 points**, **0 points**	**10 points**, **5 points**, **0 points**
Pros	High scoring	Guaranteed points, good point balance
Cons	All points rely on toggles	Wastes the red half of a red yellow pin
Other Notes	If we used this stack, we would have to really make sure we get the toggle	The yellow that could always be covered up anyway balances well with the guaranteed points from the red pin

From this analysis, it seems like the main challenge with deciding combinations of pins and cups we score during matches will be trying to balance the higher potential points of more yellow pins with getting guaranteed points by protecting reds.

Any yellow pins we score could be taken by the blue alliance with a toggle, but if we end up with the toggle, that is more points than normal reds.

Scoring ten points worth of protected red pins would take twice or more times as long as one yellow pin, but scoring those protected reds is much safer because we get guaranteed points.

If we are taking the chance and trying to score yellows, we need to consider whether we want to score a lot of them or not. This is an interesting risk vs reward scenario because there is no difference between scoring a large amount of yellows vs a small amount in risk over reward.

A large amount of the points alliances get will come from yellow pins and therefore having toggles. We need to stay aware of this and all toggles during matches.

During the endgame, it could very well be a lot more worth it to go for toggles than to try to get the midfield. We would only need to have two yellow pins scored in a single quadrant for going for the toggle to be more points than both robots in the midfield.

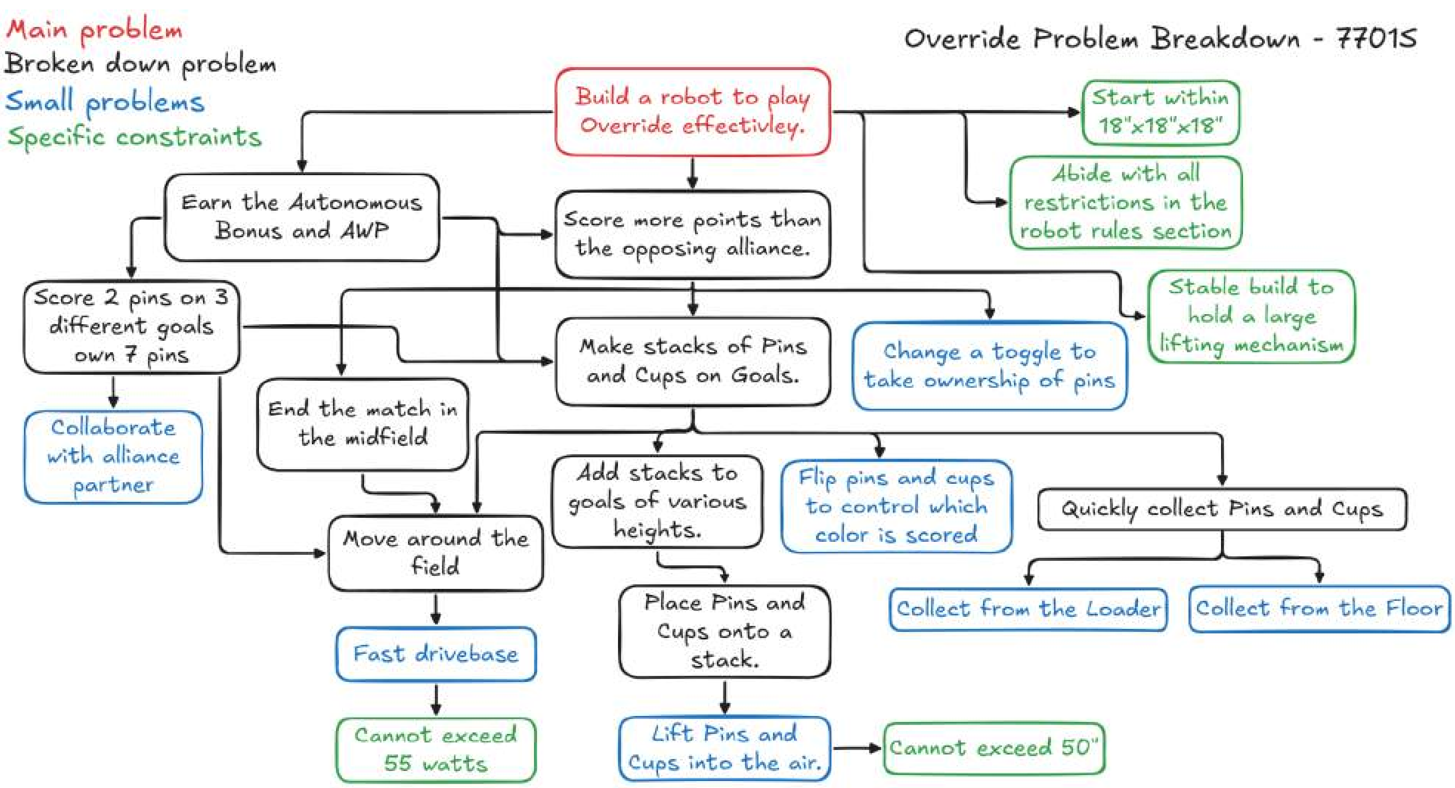
We have identified that scoring guaranteed points is also good and something we should do during matches because of how easy it would be to take away non guaranteed points (covering pins, switching toggles, even just ending the match touching a toggle).

From the above analysis, YRC is the best match load combination for scoring guaranteed points. For this, the opaque side of the cup is facing down, so that means we don't need to worry what pin we score it on, and could even put it on an exposed blue pin to descore it.

Any of the combinations with the clear side of the cup facing down can be placed on red pins in goals to get guaranteed points, but that is more work.

Problem Breakdown Chart

To better understand all of our design goals, we made a chart to break down the main problem (playing Override) into smaller problems. These smaller problems are what we need to design our robot around to accomplish the main goal. The light blue boxes are the base problems we need to design our robot around. The arrows signify breaking down problems, and what problems we need to solve in order to do something else (in order to place pins on a stack, we need to lift them into the air).



Physical Field Analysis

Members Present	EDP Steps	Problems or Goals	Scheduling
Liam Shipman	Identify the Problem	Analyze the physical attributes of the game elements to better understand a robot concept.	Completed Date: Apr 25, 2026 - Apr 26, 2026

As VEX has just begun shipping fields, not many teams have them yet, and most people have only been able to assume things about the field based on the game reveal and the objects given out after the reveal. We just received ours, and wanted to take a look to find more information about the elements.

Pins, Cups, and Goals

The goals are mostly hollow shells, supported by standoffs, which are connected to metal plates. After the game was revealed, people thought stacks would be extremely easy to descore. However, when you actually put a stack inside a goal, it is actually extremely difficult to descore unless you apply upward force. This is very reassuring, and we don't have to be nearly as careful as we initially thought. You can hit the tower from the side and curve it significantly before it falls over.

Also, alliance goals can hold stacks of 7 pins before it gets taller than 50”, and neutral and center goals can only hold stacks of 6 pins. Finally, every goal on a certain half of the field has an original April tag (mirrored across).

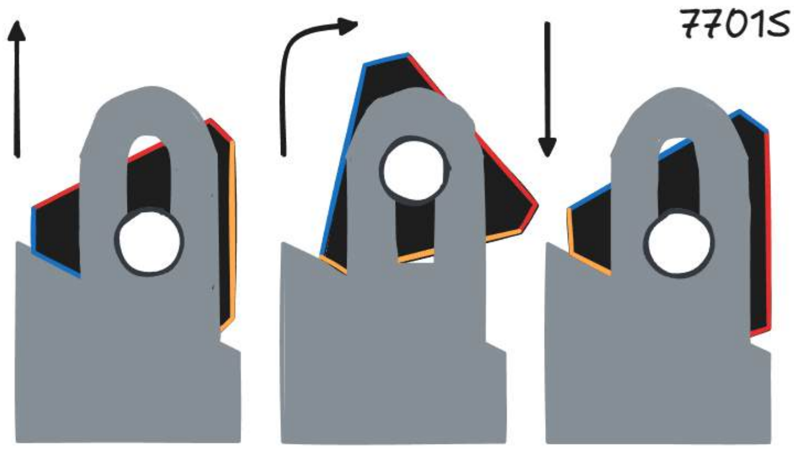
Showing how you can push a goal significantly before it collapses.

Toggles

We had seen in the trailer that the toggles were on a linear slide, but we weren't exactly sure how they worked or how easy they would be to spin. We found they spin quite easily and lift as they rotate (due to their shape and the linear slide). They also snap into place, and it is very difficult to have them rest between colors on the corner of the triangle.

A picture showing the toggle linear slide mechanism.

The toggle spins because of its linear slide. As you rotate it, it rises on the slide as the curve of the triangle pushes on the lower surface until it reaches the apex/corner of the triangle. Then, as you continue rotating, it lowers back, snapping into place.



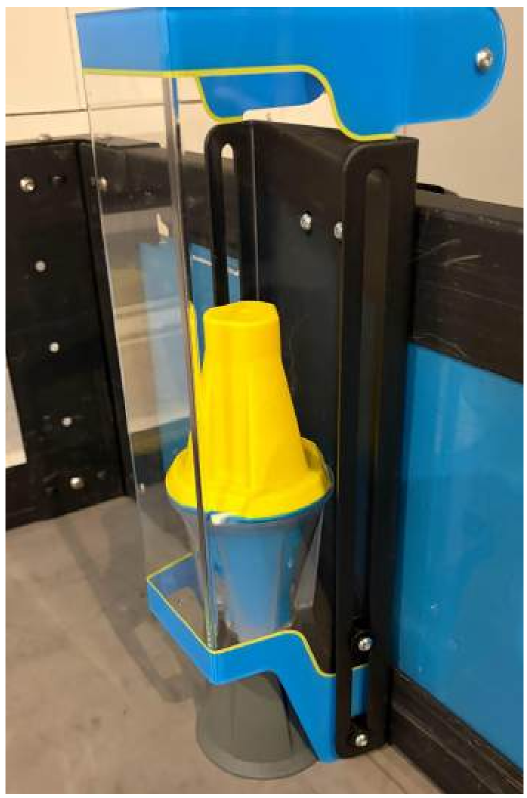
A diagram showing how the toggles function, made by 7701S 5/1/26.

Loaders

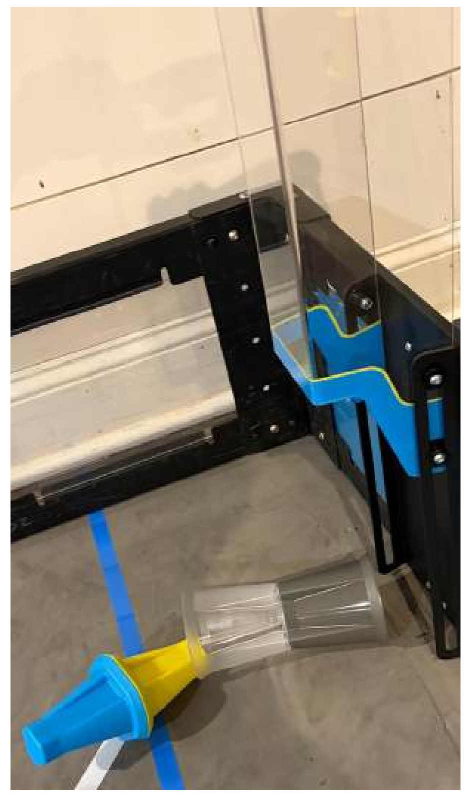
The loaders can lift to allow robots to collect them. There is a handle to help the person to lift it, and the lifting section is pretty light.

Also, you can load in two ways: First, you can drop them in from the top, keeping them stacked. Second, you can load it from the back when it is lifted. However, this causes the pin and cup to fall over and break away from each other.

This can be beneficial as you can load to accommodate your robot's design (standing up/fallen).



Showing when you load from the top. The pins and cup are standing up.



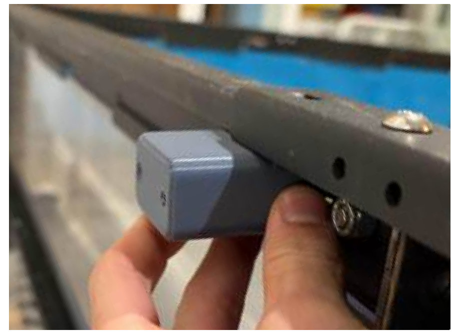
Showing when you load from the back when lifted. The pins and cup fell over.

Field Issues

The field we got and built was a personal field for our team at Liam's house, not the school fields (probably won't arrive until June-July). Liam has an old, metal field perimeter, not a new white one. This caused problems as the game was designed for the new portable perimeters. This caused the toggles and loaders to not fit correctly and would require spacers to fix. Since we did not have spacers available, we opted to 3D-print spacers of the correct size.

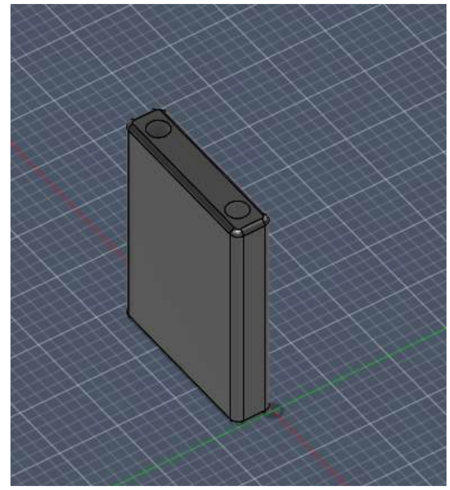
Showing the toggles not designed for an old field as it doesn't fit correctly and is hanging down.

Liam designed them in Fusion after measuring the dimensions of the bracket that holds the toggles up and the field. The original design was too wide, so we remade it so it was much more sleek, and it fit well. Once we printed it, Liam shared the file publicly so other teams could use it if they only had old perimeters like us.


Showing the first design, which was too wide and whose holes didn't align correctly.


Showing the second iteration with a toggle. It was much more sleek and fit very well.


Showing the second iteration on the loaders.


Showing the second iteration in the CADs.

Design Brief - P.1.0.0

Members Present	EDP Steps	Problems or Goals	Scheduling
Liam Shipman Sam Wilson	Identify the Probl...	Explain the mechanisms and design criteria for our first robot.	Completed Date: May 2, 2026 - May 3, 2026

Problem Statement

Our robot needs to be able to score as many points as possible while working with another robot. It could score points by making stacks of pins and cups, taking ownership of pins by flipping toggles, and moving to the midfield by the end of the match.

Design Statement

We will design, build, test, and program a robot to complete the tasks of the game. It needs to be able to score cups and pins, take control of toggles, move quickly and effectively around the field, move accurately on a programmed path, and be strong and durable during competition. We will use the design process to iterate and optimize our robot design throughout the season.

Based on our analysis and what problems our robot needs to solve, we have determined what mechanisms we need to make based on our initial strategy. We then made design criteria for each specific mechanism, which should help narrow down solutions in the future.

Constraints

- Starts in an 18”x18”x18” footprint, can not expand past 24”x24”x50”.
- 2 air tanks.
- Needs to be light; if it's heavy, it is inefficient.
- Only physical modifications to most parts.
- 12 4”x8” pieces of custom plastic.
- No 3D-printed parts.
- Needs to be strong and sturdy to withstand the rigor of competition.
- 88 watts of motor power total, maximum of 55 watts of motor power on the drivetrain.
- 1 brain, 1 battery, no electronic modifications.

Design Criteria

1. **Drivetrain**
   a. It should be fast to allow us to quickly cross the field at a moment's notice.
   b. It should use all 55 watts so we can guarantee that we are not weaker than other robots, and that we can compete in the midfield at the end of the match.
   c. It should be quite durable to withstand the rigor of competition and connect the rest of the robot together.
   d. While being fast, it should still allow for precise movements (especially when scoring high stacks).
   e. It should have a low center-of-gravity to prevent tipping.
   f. It should be light to increase its efficiency (the lighter the robot is, the higher the performance the motors have).

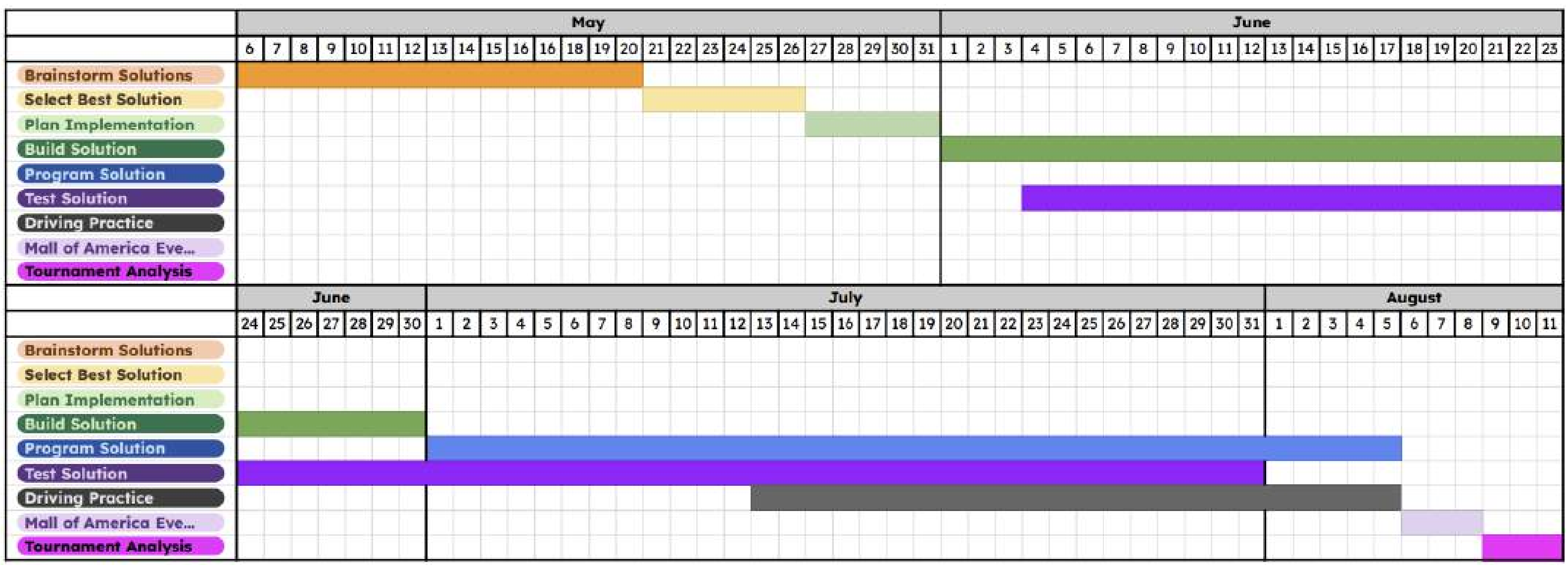
2. **Toggle Spinner**
    a. It should be able to quickly change the toggle with relative ease.
    b. It can use a motor if necessary, but being passive is preferred.
    c. It should be somewhat light and not shift the overall weight significantly.
3. **Intake**
    a. It should be able to intake pins and cups from multiple orientations.
    b. It should be able to collect every combination from the match loader.
    c. It should be able to rotate the pins/cups so you can control which side gets scored.
    d. It should be fast, strong, and reliable.
    e. It should be lightweight to prevent tipping and increase drivetrain efficiency.
4. **Lift**
    a. It should be able to lift pins/cups to various heights.
    b. It should always maintain legal size limits (start and end at 18”, go up to 50”).
    c. It should be quite fast so we don’t spend too much time waiting on the lift when we could be doing other things.
    d. It should be durable and withstand hits from other robots.
    e. It should be light to improve overall efficiency and reduce tipping risk.
    f. It should be traceable with PID movements, so we always know its position.
5. **Miscellaneous**
    a. The robot should have some sort of anti-tip, so if we do fall over, we can still recover.
    b. The robot should include various sensors so we can detect the color of the toggle and potentially detect the AprilTags.
    c. The robot should include an aligner for goals to help with scoring consistency and speed of lining up.

Timeline - P.1.0.0

Members Present	EDP Steps	Problems or Goals	Scheduling
Liam Shipman Sam Wilson Nolan Wolff	Identify the Probl...	Determine our timeline for our first robot iteration.	In Progress - On Time Date: May 5, 2026

First Iteration Gantt Chart

We made our first Gantt chart and overall first design timeline to match each person's schedule (seen on page __) and with the Mall of America signature event, an event we would like to attend, which is in early August. Also, due to many members having AP testing and finals, May will have a lot more time and will mainly be researching different designs. Also, these dates are just general ideas; we do not need a strict guideline because of how much time we have.



This is the Gantt Chart we made. We already have most of our game analysis done at this point, so it is not mentioned on the chart. We plan to continue brainstorming over the next few weeks (until the 20th), then select solutions afterward. We hope to be creating the CAD in the last week of May so we can begin building in June. During the building, we will constantly be running tests to make sure the solution is effective. After building, we will begin programming in July (continuing testing). Finally, from mid-July to August, we also included drive practice. This schedule should hopefully give us a general timeframe for this robot and give everyone enough time with their summer plans. Also, to ensure we stay on schedule, we have an On Time/Late section of the "Scheduling" portion of the header (seen above). This will allow us to see how well we are staying on top of things.

Brainstorm Solutions - All members will come up with ideas and research so we can have a large number of possibilities to consider.
Select Best Solution - All members will weigh in on the decisions to ensure we don't leave out possibilities.

Plan Implementation - Sam is the main CADer, so it will be his job to construct the CAD. All members may give him input to influence his design.

Prog Notebook

The Basics

Members Present	EDP Steps	Problems or Goals	Scheduling
Nolan Wolff	Program the So... Team Manage...	Decide how our program will be managed and written this year.	In Progress - N/A Date: May 15, 2026 - May 16, 2026

Basic Programming Guidelines

- All programs will be written in C++ or C.
  - Fastest for the brain to read, best to use with PROS (more about this later).
- All programs will be written with PROS syntax and uploaded through the PROS operating system to the brain.
  - There are many reasons PROS is superior to VEXos, which is what we used last year.
  - More will be explained on page ( ).
- Code will be shared through GitHub and source control on VSCode.
  - More about our repository and git etiquette on page ( ).
- The program is all written by our team.
  - Use of AI is strictly prohibited for writing code.
  - I have debated using templates for things like odometry and PID control.
    - Will talk more about this in a decision matrix on the PROS page.
- All subprograms will be part of one project.
  - This is something that I have liked for the past couple of years. It is nice ot have all of the programs in the same file, so when things are changed, multiple files don’t have to be updated.
  - Better ease of use for the drive team. Makes me able to create a Brain screen UI.
- Code should be easy to read.
  - Variables will be clearly named, functions and global variables can be in camel case (ex, camelCase), and private and integer variables can be in snake case (ex, snake_case).
  - This makes it really easy to discern between variables and what they do. Sometimes this will be messed up, and that’s okay, as long as the names are descriptive.

Choosing PROS

Members Present	EDP Steps	Problems or Goals	Scheduling
Nolan Wolff	Program the So... Brainstorm Sol...	Decide what operating system we will be programming in year-round.	In Progress Date: May 15, 2026 - May 17, 2026

What is an operating system?

- An operating system, or OS, as it relates to VEX robotics, is the compilation of the program to the brain, as well as the running of the program in the brain.
- The differences between Operating systems can be very large or small.
- Over the last couple of years, I have been coding using VEXos, the default operating system that VEX provides.
- Before this season and in prior seasons, I have heard that PROS is faster, but I decided to create a decision matrix to decide if I should switch or stay.

Appendix A: Referenced Game Rules

Appendix B: Citations

Override 1.0 Game Manual, VEX Robotics,
https://content.vexrobotics.com/docs/2026-2027/override/files/override-0.1-game-manual.pdf

7701Y Push Back Notebook (Previous Season's Notebook)

Purdue ACM SIGBots. PROS for V5, version 3.8.0. Accessed May 15, 2026.
https://pros.cs.purdue.edu/

Appendix C: Old Content

1: Robot Numbering System, Push Back, 7701Y (Liam, Nolan, Sarthak and Grayson were all members of this team and made this system).

Team Management

Completed : Jun 8, 2025

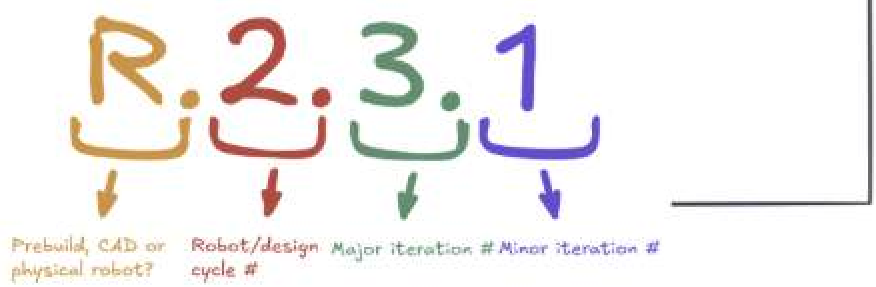
Author(s): Liam Shipman

Version P.0.0.0

Management - Numbering System

Goal: Explain the numbering format we use to show robot iteration.

Sketch explaining numbering system. Sketch by 7701Y.



Inspiration

- This format was inspired by team 2654E Echo's High Stakes notebook (can be viewed here: https://2654e.netlify.app/nb.pdf).
- We modified it to be more specific (included prebuilt and changed how the numbers increased) and to fit what we wanted from the system.

Explanation

- Letter: Is the entry or picture referring to the prebuild (P) (planning, brainstorming before building the robot/mechanism), the CAD (C), or the physical robot/build (R)?
  - In this example, we are referring to the physical build.
- First number: What full design cycle is this referring to (complete rebuild)?
  - In this example, this is the second robot of the season.
- Second number: What major design iteration is this? (ex: completely rebuilding a subsystem, adding/removing a mechanism)
  - In this example, there have been 3 major iterations.
- Third Number: What minor design iteration is this? (ex: modifying a mechanism, adding something small)
  - In this example, there has been one minor iteration.
- When we increase a number by one, succeeding numbers reset to zero (ex: P.2.1.4-> P.3.0.0, R.1.3.4->R.1.4.0).
- Why?
  - This numbering system will allow us to keep track of our current and past design iterations, which will help with referencing photos, previous designs, and should help judges understand what design we are talking about.

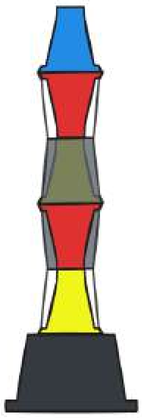
Appendix D: Extraneous Content

Extraneous stuff:

Stack notation:

- Written from top to bottom of stack (first encountered part moving downwards from the top)
- Letters for part of pin (always both halves together with top color letter first, R=red, B=blue, Y=yellow)
- Letters for cups (one letter for each orientation of cup, O=opaque on top, C=clear on top)
- G=goal (will always be on the bottom if stack is scored, since the part of a pin that is in the goal is not relevant, exclude the rest of that pin)

Example:

 This stack would be referred to as BRCYROYG but this is a more complicated stack than we would likely be using this system to refer to.

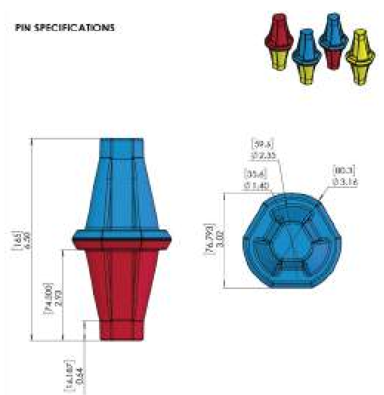
Formatting

I

Members Present	EDP Steps	Problems or Goals	Scheduling
			Not Started - N/A Date: Apr 25, 2026 - Apr 25, 2026

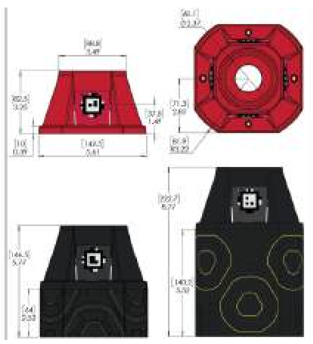
Pictures: **(crop to your liking and allat, i can get these pictures with removed backgrounds if you want just tell me which):**

- Pins:
  - Game manual pg 101, 119

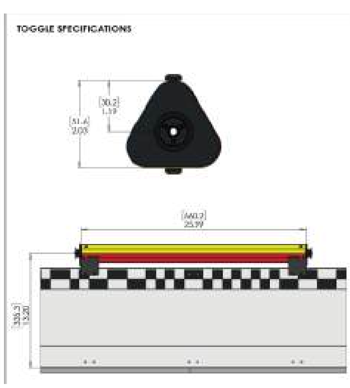


- Cups:
  - Game manual pg 102, 113

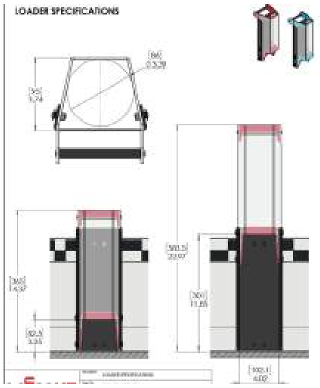
- Goals:
  - Game manual pg 103, 115



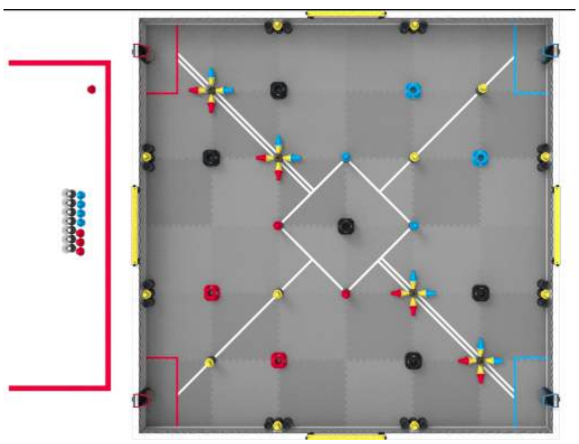
- Toggles:
  - Game manual pg 104, 121



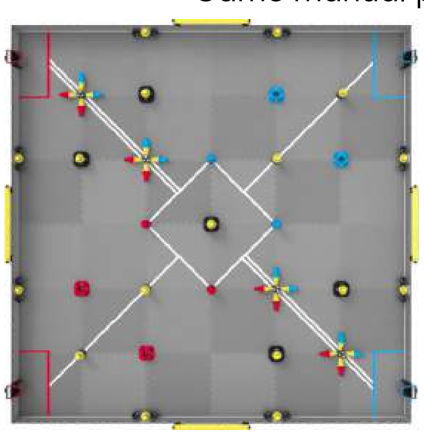
- Loaders:
    - Game manual pg 105, 116



- Skills field (to draw route on or smth)
    - Game manual pg 53



- Match field (to draw autos or other match strategy on)
    - Game manual pg 21



Max score:

(2x, 2x) Alliance goal: one pin in goal, 7 cups with opaque side down, yellow anything pin with yellow in the transparent side, a yellow yellow pin on top. (8 yellow scored with toggle)

(4x) Neutral goal: one pin in goal, 6 cups with opaque side down and yellow anything pin with yellow in the transparent side, a cup with yellow yellow pin on top (8 yellows scored with toggle)

(1x) Midfield goal: one pin in goal, 6 cups with opaque side down and yellow anything pin with yellow in the transparent side but yellow yellow pin on top. (7 yellows scored with toggle)

Resources

https://excalidraw.com/#json=EDXJNlQp7M0yQNmNtgu3H,3pjy85WZQ9HWFFn7pr1EfA

Rubrics

Notebook

Engineering Notebook Rubric (Page 1 of 2)

Team # ___________ **Grade Level ☐ ES | ☐ MS | ☐ HS | ☐ University Judge Name** ____________________

Directions: Determine the point value that best characterizes the content of the Engineering Notebook for that criterion. Write that value in the column to the right. This rubric is to be used for all Engineering Notebooks regardless of format (physical or digital). Please refer to Section 5 of the Guide to Judging for information on how to use this rubric.

Note: Any student-centered or academic honesty concerns, such as plagiarism, should be brought to the attention of the Judge Advisor and/or Event Partner.

CRITERIA	PROFICIENCY LEVEL			
ENGINEERING DESIGN PROCESS	**EXPERT** (4-5 POINTS)	**PROFICIENT** (2-3 POINTS)	**EMERGING** (0-1 POINTS)	**POINTS**
IDENTIFY THE PROBLEM / DESIGN GOAL(S)	Clearly identifies the problem / design goal(s) in detail at the start of each design process cycle. This can include elements of game strategy, robot design, or programming, and should include a clear definition and justification of the design goal(s), criteria, and constraints.	Identifies the problem / design goal(s) at the start of each design cycle but is lacking details or justification.	Does not identify the problem / design goal(s) at the start of each design cycle.	____
BRAINSTORM SOLUTIONS	Explores several different solutions with explanation. Citations are provided for ideas that came from outside sources such as online videos or other teams.	Explores few solutions. Citations provided for ideas that came from outside sources.	Does not explore different solutions or solutions are recorded with little explanation.	____
SELECT BEST SOLUTION	Fully explains the "why" behind design decisions in each step of the design process for all significant aspects of a team's design.	Inconsistently explains the "why" behind design decisions.	Minimally explains the "why" behind design decisions.	____
BUILD AND PROGRAM THE SOLUTION	Records the steps the team took to build and program the solution. Includes enough detail that the reader can follow the logic used by the team to develop their robot design, as well as recreate the robot design from the documentation.	Records the key steps to build and program the solution but lacks sufficient detail for the reader to follow their process.	Does not record the key steps to build and program the solution.	____
ORIGINAL TESTING OF SOLUTIONS	Records all the steps to test the solution, including test results. Testing methodology is clearly explained, and the testing is done by the team. Original testing results are explained and conclusions are drawn from that data.	Records the key steps to test the solution. Testing methodology may be incomplete, or incomplete conclusions are recorded.	Does not record steps to test the solution. Testing or results are borrowed from another team's work.	____
REPEAT DESIGN PROCESS	Shows that the design process is repeated multiple times to work towards a design goal. This includes a clear definition and justification of the design goal(s), its criteria, and constraints. The notebook shows setbacks that the team learned from, and shows design alternatives that were considered but not pursued.	Design process is not often repeated for design goals or robot/game performance. The notebook does not show alternate lines of inquiry, setbacks, or other learning experiences.	Does not show that the design process is repeated. Does not show setbacks or failures, or seems to be curated to craft a narrative.	____

NOTES:

Engineering Notebook Rubric (Page 2 of 2)

ENGINEERING NOTEBOOK FORMAT AND CONTENT	EXPERT (4-5 POINTS)	PROFICIENT (2-3 POINTS)	EMERGING (0-1 POINTS)	POINTS
INDEPENDENT INQUIRY	Team shows evidence of independent inquiry from the beginning stages of their design process. Notebook documents whether the implemented ideas have their origin with students on the team, or if students found inspiration elsewhere.	Team shows evidence of independent inquiry for some elements of their design process. Ideas and information from outside the team are documented.	Team shows little to no evidence of independent inquiry in their design process. Ideas from outside the team are not properly credited. Ideas or designs appear with no evidence of process.	___
USABILITY & COMPLETENESS	Records the entire design and development process with enough clarity and detail that the reader could recreate the project's history. Notebook has recent entries that align with the robot the team has brought to the event.	Records the design and development process completely but lacks sufficient detail. Documentation is inconsistent with possible gaps.	Lacks sufficient detail to understand the design process. Notebook has large gaps in time, or does not align with the robot the team has brought to the event.	___
ORIGINALITY & QUALITY	Cited content is kept to relevant information and all cited content longer than a paragraph is located in appendices to the Engineering Notebook. Information originating from outside the team is always properly cited in the notebook with the source and date accessed. Most or all Engineering Notebook content is original to the submitting team members.	Cited content is mostly kept to relevant information. Information originating from outside the team is properly credited. Cited content is paraphrased with some original content describing the team's design process.	Cited content is excessive and/or is not kept in appendices, or non-original content is not cited. Plagiarised content should be noted to the JA and through the REC Foundation Code of Conduct process.	___
ORGANIZATION / READABILITY	Entries are logged in a table of contents. There is an overall organization to the document that makes it easy to reference, such as color coded entries, tabs for key sections, or other markers. Notebook contains little to no extraneous content that does not further the engineering design process.	Entries are logged in a table of contents. There is some organization to the document to enhance readability. Notebook contains some extraneous content that does not further the design process, but it does not severely impact readability.	Entries are not logged in a table of contents, and there is little adherence to a system of organization. Excessive extraneous content makes the notebook difficult to read, use, or understand.	___
RECORD OF TEAM & PROJECT MANAGEMENT	Provides a complete record of team and project assignments; contains team meeting notes including goals, decisions, and building/programming accomplishments; design cycles are easily identified. Resource constraints including time and materials are noted throughout. Notebook has evidence that documentation was done in sequence with the design process. Entries include dates and names of contributing students.	Records most of the information listed at the left. Level of detail is inconsistent, or some aspects are missing. There are significant gaps in the overall record of the design process. Notebook may have inconsistent evidence of dates of entries and student contributions.	Does not record the design process in a way that shows team progress. There are significant gaps or missing information for key design aspects. Notebook has little evidence of dates of entries and student contributions.	___
INNOVATE AWARD NOTES (optional):				TOTAL POINTS ___

All judging materials are strictly confidential. They are not shared beyond the Judges and Judge Advisor and shall be destroyed at the end of the event.

Interview

Team Interview Rubric

Team # ____________ **Grade Level** ☐ ES | ☐ MS | ☐ HS | ☐ University **Judge Name** ____________________

Directions: Determine a point value that best characterizes the content of the Team Interview for that criterion.

CRITERIA	PROFICIENCY LEVEL EXPERT (4-5 POINTS)	 PROFICIENT (2-3 POINTS)	 EMERGING (0-1 POINTS)	POINTS
ENGINEERING DESIGN PROCESS *All Awards*	Team shows evidence of independent inquiry from the beginning stages of their design process. This includes brainstorming, testing, and exploring alternative solutions.	Team shows evidence of independent inquiry for some elements of their design process.	Team shows little to no evidence of independent inquiry in their design process.	____
GAME STRATEGY *Design, Innovate, Create, Amaze*	Team can fully explain their entire game strategy including game analysis.	Team can explain their current strategy with limited evidence of game analysis.	Team did not explain game strategy, or strategy is not student-directed.	____
ROBOT DESIGN *Design, Innovate, Build Create, Amaze*	Team can fully explain the evolution of their robot design to the current design.	Team can provide a limited description of why the current robot design was chosen, but shows limited evolution.	Team did not explain robot design, or design is not student-directed.	____
ROBOT BUILD *Innovate, Build, Create, Amaze*	Team can fully explain their robot construction. Ownership of the robot build is evident.	Team can describe why the current robot design was chosen, but with limited explanation.	Team did not explain robot build, or build is not student-directed.	____
ROBOT PROGRAMMING *Design, Innovate, Think, Amaze*	Team can fully explain the evolution of their programming.	Team can describe how the current programs work, but with limited evolution.	Team did not explain programming, or programming is not student-directed.	____
CREATIVITY / ORIGINALITY *Innovate, Create*	Team can describe creative aspect(s) of their robot with clarity and detail.	Team can describe a creative solution but the answer lacks detail.	Team has difficulty describing a creative solution or gives minimal response.	____
TEAM AND PROJECT MANAGEMENT *All Awards*	Team can explain how team progress was tracked against an overall project timeline. Team can explain management of material and personnel resources.	Team can explain how team progress was monitored, and some degree of management of material and personnel resources.	Team cannot explain how team progress was monitored or how resources were managed.	____
TEAMWORK, COMMUNICATION, PROFESSIONALISM *All Awards*	Most or all team members contribute to explanations of the design process, game strategy, and other work done by the team.	Some team members contribute to explanations of the design process, game strategy, and other work done by the team	Few team members contribute to explanations of the design process, game strategy, and other work done by the team.	____
RESPECT, COURTESY, POSITIVITY *All Awards*	Team consistently interacts respectfully, courteously, and positively in their interview.	Team interactions show signs of respect and courtesy, but there is room for improvement.	Team interactions lack respectful and courteous behavior.	____
SPECIAL ATTRIBUTES & OVERALL IMPRESSIONS *Judges, Inspire*	Does the team have any special attributes, accomplishments, or exemplary effort in overcoming challenges at this event? Did anything stand out about this team in their interview? Please describe:			TOTAL POINTS ____

All judging materials are strictly confidential. They are not shared beyond the Judges and Judge Advisor and shall be destroyed at the end of the event.

Team Interview Notes

Directions: Use this sheet to take notes during each Team Interview. As a Judge group, ask open ended questions to teams that give insight into each of the criteria below.

Team Number ___________ **Judge Name** ____________________

CRITERIA	CRITERIA EXPLANATION	JUDGE'S NOTES
ENGINEERING DESIGN PROCESS *All Awards*	How well does the team explain the process they used to create their robot design?	
GAME STRATEGY *Design, Innovate, Create, Amaze*	Can the students explain their game strategy, how they came up with it, & how well it fits with their robot design?	
ROBOT DESIGN *Design, Innovate, Create, Amaze, Build*	Do students demonstrate ownership of the design process? Is the robot well designed to accomplish their goals?	
ROBOT BUILD *Innovate, Build, Create, Amaze*	Do students demonstrate ownership of the build process? Is the robot well-built and robust?	
CREATIVITY / ORIGINALITY *Innovate / Create*	Does team describe creative aspect(s) of their robot with clarity and detail?	
ROBOT PROGRAMMING *Think*	Do students demonstrate ownership of the robot's programming? How well can they explain their code?	
TEAM & PROJECT MANAGEMENT *All Awards*	Can students explain how they managed their time, resources, and people to work effectively?	
TEAMWORK, COMMUNICATION, PROFESSIONALISM *All Awards*	Do all team members share in the work of being a successful team? Does everyone contribute in some way?	
RESPECT, COURTESY, POSITIVITY *All Awards*	Did students answer respectfully and courteously?	
SPECIAL ATTRIBUTES *Judges, Inspire*	Does the team have any special attributes or accomplishments?	

All judging materials are strictly confidential. They are not shared beyond the Judges and Judge Advisor and shall be destroyed at the end of the event.